

Shaan Pakala

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[Google Scholar](#), [LinkedIn](#) & [GitHub](#)

About

I am a 2nd year Computer Science Ph.D. student at the University of California, Riverside, where I work on AI4Science research problems with my advisor Professor [Vagelis Papalexakis](#). This includes developing interpretable surrogate models for designing new materials, or efficiently generating expensive physics simulation data using generative machine learning.

Education

University of California, Riverside

- Ph.D. in Computer Science
- B.S. in Data Science

Sept. 2025 – Present
Sept. 2021 – June 2025

Research Experience

Lawrence Livermore National Laboratory

- **Research Intern** **Summer 2026**
Intern in the predictive healthcare group, with Dr. Priyadip Ray. I worked on using machine learning for early disease outbreak detection, using the Tuolumne supercomputer.
- **Research Intern** **Summer 2025**
Intern in the predictive healthcare group, with Dr. Priyadip Ray and Dr. Braden Soper. I worked on causal inference for Parkinson's Disease progression, with clinicians/biomedical researchers.

University of California, Riverside

- **Graduate Student Researcher** **June 2025 – Present**
I work with domain-experts in physics & material science on AI4Science problems using tensor decomposition: (i) surrogate modeling for robust & interpretable material design [1] [3] and (ii) efficient physics simulation generation using tensor decomposition in GANs & diffusion models [4].
- **Undergraduate Student Researcher** **June 2024 – June 2025**
Research with Prof. Papalexakis. I worked on automating the optimization of data science pipelines (e.g., neural network architecture search, SQL queries) with tensor completion [2].

Papers (* denotes equal contribution)

Conference

- [1] **Tensor Methods: A Unified and Interpretable Approach for Material Design**
[Shaan Pakala](#), A.E. Gongora, B. Giera, and E.E. Papalexakis
Proceedings of ACM SIGKDD (AI for Sciences track) 2026 [PDF] [Code]
- [2] **Automating Data Science Pipelines with Tensor Completion**
[Shaan Pakala](#), B. Graw, D. Ahn, T. Dinh, M.T. Mahin, V. Tsotras, J. Chen, and E.E. Papalexakis
Proceedings of IEEE International Conference on Big Data 2024 [PDF] [Code]

Workshop

- [3] **Surrogate Modeling for the Design of Optimal Lattice Structures using Tensor Completion**
Shaan Pakala, A.E. Gongora, B. Giera, and E.E. Papalexakis
NeurIPS AI for Accelerated Material Design Workshop 2025 [[PDF](#)] [[Code](#)]
- [4] **Efficiently Generating Multidimensional Calorimeter Data with Tensor Decomposition Parameterization**
P. Goulart*, Shaan Pakala*, and E.E. Papalexakis
ICCV Representation Learning with Very Limited Resources Workshop 2025 [[PDF](#)] [[Code](#)]
- [5] **Tensor Completion for Surrogate Modeling of Material Property Prediction**
Shaan Pakala, D. Ahn, and E.E. Papalexakis
AAAI Bridge on Knowledge-Guided Machine Learning 2025 [[PDF](#)]

Awards

- **Travel Award** (ACM KDD'26, NeurIPS'25 AI4Mat Workshop, IEEE BigData'24)
- **Undergraduate Research Spotlight Award** (UC Riverside)
- **NSF Research Experience for Undergraduates** (UC Riverside)

Contributed Talks & Posters

- **Talk**, LLNL CASIS Workshop, “Tensor Methods for Material Design,” 2026
- **Talk**, UTRGV MECIS Spring Webinar, “Surrogate Modeling with Tensor Completion,” 2026
- **Poster**, SIAM SDM Ph.D. Forum, “Exploring Design Spaces with Tensor Decomposition,” 2026
- **Poster**, UCR Detector Science Workshop “Tensor Methods for Interferometer Simulations,” 2026
- **Poster**, SAIR UCR Science & AI Summit “Tensor Methods for Material Design,” 2026

Academic Service

- **Organizer**: [AI4AutoSci](#) Workshop at IEEE BigData'26
- **Reviewer**: Journal of Machine Learning Research, TensorKDD Workshop at KDD'26, AI Data Scientist Workshop at KDD'26